

Abstention and Refusal Are Separable Directions in the Residual Stream

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Abstract

Instruction-tuned language models refuse harmful requests and can abstain when the context is unreliable or unrelated to the factual question. The two responses look alike on the surface, and prior work has not tested whether they share the residual-stream direction that mediates refusal. We test this with a cross-intervention at matched benign-refusal cost across three instruction-tuned model families. In one family, steering the abstention direction moves abstention while steering the refusal direction moves it by zero. The recipe is not deployable because it fails its accuracy budget and does not transfer outside its training distribution, but the dissociation means that tools built for refusal should not be repurposed for conflict abstention without an explicit cross-intervention check.

1 Introduction

A language model that answers a factual question under a given context can produce three outcomes. It can answer from its own knowledge, follow the context, or abstain. Prior work characterises the first two outcomes in detail (Jin et al., 2024; Zhao et al., 2024; Islam et al., 2025), while the third is typically treated as a non-answer. It is also the desirable response when the context is corrupted, outdated, or unrelated to the question.

A parallel line of work on safety alignment shows that refusal of harmful prompts is mediated by a single direction in the residual stream (Arditi et al., 2024). The residual stream is the running activation that each transformer layer reads from and writes to. Whether the same direction also mediates abstention under unreliable context is open. If the direction is shared, the safety-direction toolkit applies directly to conflict abstention. If it is not, methods that approximate abstention by probing or steering the refusal direction misattribute the signal. Geometry alone is insufficient evidence, because

two directions can be linearly distinct and still drive the same downstream computation. We therefore use a cross-intervention test, asking whether the direction that decodes cautious abstention is also the one that moves it when steered at a benign-refusal cost matched to the refusal direction.

Related work. Three lines of prior work bear directly on this question. Knowledge-conflict studies characterise model behaviour when the model’s stored knowledge disagrees with a passage (Longpre et al., 2021; Xie et al., 2024; Xu et al., 2024), but they treat abstention as a downstream outcome rather than a target representation. Refusal-direction work has refined the geometry, showing that the direction occupies a small subspace rather than a single vector (Wollschläger et al., 2025), that harmfulness and refusal are themselves separable (Zhao et al., 2025), and that harmful-refusal and over-refusal occupy distinct subspaces with the latter task-conditioned rather than global (Maskey et al., 2026). Those partitions sit inside the safety-refusal family. We test a different pair, between safety refusal and conflict-induced abstention, and we ground it in a causal cross-intervention rather than probe separation alone. A separate lineage shows that hidden states carry signals about what a model knows and does not know (Kadavath et al., 2022; Yin et al., 2023; Slobodkin et al., 2023), and behavioural fine-tuning teaches models to say so directly (Zhang et al., 2024). Conflict abstention sits at the intersection of these three lines and has not been examined as a representation in its own right.

Contributions. We make three contributions. (i) Across three instruction-tuned families a linear classifier reads cautious-framing abstention from residual states, and the recovered direction sits close to orthogonal to the canonical refusal direction. (ii) On Gemma, at a benign-refusal cost matched to the refusal direction, the abstain mean-difference

vector moves conflict abstention while the refusal direction does not. (iii) The recipe does not transfer to two out-of-distribution short-answer benchmarks at the same nominal scale on any of the three families, so the representation generalises while the intervention does not.

Implications. Two practices follow. Probes that read abstention off the refusal direction misattribute the signal in at least the regime we tested, and methods that try to elicit abstention by amplifying refusal do not work there either. Conflict abstention should be treated as a distinct target representation with its own extraction and its own evaluation, rather than as a byproduct of safety machinery.

2 Design and Method

Knowledge-conflict data. We sample PopQA (Mallen et al., 2023), a short-answer factual question-answering (QA) set built from entity-relation triples, and retain 426 of 1500 items for which Gemma-2-9B-IT returns the same gold answer under three independent no-context generations (mean consistency 0.962). For 250 retained facts we build four context cells, one with a false counterfactual context, one with a true context, one with an irrelevant context, and one with no context. Each item is rendered under three framings, asking the model to use its own knowledge, to use only the context, or to answer cautiously. Greedy generations are labelled by a regex-and-alias rule into PK-WIN (parametric-knowledge win: the model answers correctly from its own knowledge), CK-WIN (context-knowledge win: the model follows the false context), ABSTAIN, or OTHER, yielding 3000 prompts.

Directions. For each model we extract last-token residuals at every layer. The abstain probe is a per-layer logistic regression on the cautious-framing subset (946 rows, four context cells), scored by area under the ROC curve (AUC) on a held-out test split with 95% confidence intervals (CIs). Per-cell composition is uneven and most cautious abstentions arise under irrelevant context (Tables 4 and 5, Appendix A), so the headline decoding is of cautious-framing abstention across cells rather than abstention isolated to the counterfactual cell. The refusal direction ρ_ℓ is the mean difference between harmful and benign JailbreakBench prompts (Chao et al., 2024) following Arditi et al. (2024). We steer by adding a scaled abstain mean-difference

vector to the last-token residual-stream activation at layer ℓ , following activation addition (Turner et al., 2024). Let $h_\ell \in \mathbb{R}^d$ denote that activation, and $\mu_{\text{abs},\ell} = \bar{h}_\ell^{\text{abs}} - \bar{h}_\ell^{\neg\text{abs}}$ the difference between the mean residual on cautious-framing abstain rows and the mean residual on cautious-framing non-abstain rows. The intervention is

$$h_\ell \leftarrow h_\ell + s \mu_{\text{abs},\ell}, \quad (1)$$

with s in raw activation units of $\mu_{\text{abs},\ell}$. We steer at $\ell = 20$, the injection layer set in our pilot and held fixed across families to avoid post-hoc layer selection. Three gates set in the pilot apply. Conflict abstention should rise by at least $\Delta = 20$ percentage points (pp), factual loss should stay at or below 5 pp, and benign false-positive (FP) rate should stay at or below 5 pp.¹

3 Results

Abstention is linearly decodable. Trained on the cautious-framing subset to separate abstain rows from the other label classes, the probe on Gemma peaks at layer 34 with test AUC 0.980 (Figure 2). The same probe trained per family reaches AUC 0.955 on Llama-3.1-8B-Instruct and 0.979 on Qwen-2.5-7B-Instruct (Figure 1), so the decoding result is not specific to one architecture. Decoding survives at AUC 0.973 when irrelevant-context rows are excluded (Appendix A, variant B), so the headline decoder reads abstention rather than irrelevant context.

Labels survive a two-judge audit. A second labelling pass uses Gemma and Qwen as independent string-level judges on 800 cautious-framing rows stratified at 200 per cell. The two judges agree on 90.1% of rows with Cohen’s $\kappa = 0.842$. Regex precision against the two judges is 1.00 and 0.990, and only 2 of 191 regex-abstain rows are rejected by either judge. Regex recall is 0.749 and 0.797, so the labels are conservative rather than inflated. Re-evaluated against judge labels, the layer-34 probe drops from 0.980 to 0.931 against the Gemma judge and 0.969 against the Qwen judge (Table 1 caption), and stays well above chance.

¹The $\Delta = 20$ pp threshold was set against a single-seed pilot of +32.5 pp on PopQA, before the five-seed runs. The five-seed mean in Table 2 is +4.0 pp, and we treat the pilot as noise-inflated. Because the original cloud artifacts were lost, our pre-registration memo is a reconstruction that records the referenced freeze-state commit SHA rather than an external pre-registration.

Model	Abs. L	AUC	Ref. L	$ \cos $
Gemma-2-9B-IT	34	0.980	17	0.048
Llama-3.1-8B	14	0.955	25	0.021
Qwen-2.5-7B	18	0.979	17	0.023

Table 1: Cross-family summary. AUC is the cautious-only abstain-versus-rest test AUC against the regex label pipeline at the best probe layer. For Gemma the same probe scores AUC 0.931 (95% CI [0.901, 0.956]) against the Gemma judge and 0.969 [0.948, 0.986] against the Qwen judge, with Cohen’s $\kappa = 0.842$ between judges. Llama and Qwen are not judge-audited. $|\cos|$ is the same-layer cosine between the abstain probe direction and the refusal direction.

The abstain direction is geometrically dissociated from refusal and from correctness. For Gemma-2-9B-IT (Gemma Team, 2024), the cautious-only probe direction at layer 34 has same-layer cosine 0.048 with the refusal direction, against the isotropic-null baseline $\mathbb{E}[|\cos|] = \sqrt{2}/(\pi d) \approx 0.013$ in this dimensionality (Figure 2, right). Llama and Qwen reproduce the regime (Table 1). A question-only PopQA correctness probe (Moreno Cencerrado et al., 2025) reaches test AUC 0.826 and has maximum layerwise cosine 0.064 with the abstain mean-difference, so the abstain direction is not a relabelling of confidence.

Causal asymmetry at matched benign-FP budget. Gemma abstain-direction steering at layer 20 with $s = +2$ (Equation 1) raises conflict abstention by +4.0 pp (95% CI [+3.1, +4.9]) at +1.5 pp benign-refusal cost. Refusal-direction steering at layer 17 at matched benign-FP cost yields zero conflict-abstain shift in all five seeds (Table 2, Figure 3). Pushing refusal scale beyond the matched budget moves abstention by less than one point at large benign cost. On 400 paired conflict rows, abstain steering creates 16 new abstentions while refusal steering creates none, and the paired bootstrap CI excludes zero. The dissociation rests on the asymmetry at matched cost rather than on the magnitude of the abstention shift, and is established against the canonical refusal extractor.

Domain fragility of the steering recipe. The intervention fails the factual-loss gate (+16.0 pp loss against a ≤ 5 pp ceiling) and does not transfer to SciQ (Welbl et al., 2017), a crowdsourced science multiple-choice benchmark, or NQ-open (Lee et al., 2019), the open-domain short-answer subset of Natural Questions (Kwiatkowski et al., 2019), under the same nominal scale (Table 3).

Direction	Cf. abs. Δ	Fact. loss	Ben. FP
Abs. $L20, s=+2$	+4.0 [3.1,4.9]	+16.0 [13.5,18.5]	+1.5 [0.6,2.4]
Ref. $L17, s=+2$	+0.0 [0.0,0.0]	+15.8 [12.9,18.6]	+1.8 [0.5,3.0]
Ref. $L17, s=+3$	+0.8 [0.1,1.4]	+34.5 [30.1,38.9]	+24.3 [21.3,27.2]

Table 2: Five-seed matched-budget steering summary on Gemma in percentage points with normal-approximation 95% CIs over seeds. The abstain direction has selective conflict-abstention return at matched benign-FP cost but fails the factual-loss gate. Matched-budget steering is reported on Gemma only. Llama’s near-ceiling baseline conflict abstention removes steering headroom, and a matched-budget Qwen extraction is left to future work.

Model	Domain	Base	Abs. Δ	Ref. Δ	Verdict
Gemma	SciQ	0.0	+0.0	+0.0	no transfer
Gemma	NQ-open	11.7	+1.7	+2.2	non-selective
Llama	SciQ	38.3	+22.2	+16.7	non-selective
Llama	NQ-open	92.2	−1.1	+0.6	no headroom
Qwen	SciQ	8.3	−2.2	+1.7	no transfer
Qwen	NQ-open	6.7	+0.0	+1.7	no transfer

Table 3: Out-of-domain steering at the same nominal scale, averaged over three dataset seeds, $n = 3 \times 60$ per row. Values are percentage points. Across both datasets the abstain-direction recipe either fails to move abstention or moves it together with refusal, so the intervention is PopQA-specific even though the representation is not.

Gemma and Qwen show no reliable shift on either out-of-domain slice. Llama shows a +22.2 pp mean SciQ shift under abstain steering, but refusal also moves strongly by +16.7 pp, so selectivity does not survive the domain change. The representation result is broader than the intervention. Abstention is decodable across all three regimes, and the negative transfer is consistent with the general out-of-distribution fragility of mean-difference steering vectors reported by Tan et al. (2024) and with the broader decodability-without-steerability gap documented in a separate failure regime by Liu (2026).

4 Discussion and Conclusion

Do conflict-induced abstention and safety refusal share a single residual-stream direction? The Gemma cross-intervention argues against it. At matched benign-refusal cost, steering the abstain direction produces a small conflict-abstention shift while steering the refusal direction produces none. This is stronger than any cosine argument alone, and shows that refusal-direction tooling cannot be repurposed for conflict abstention without an explicit cross-intervention check. The feature is also not a calibration-style *model knows what it*

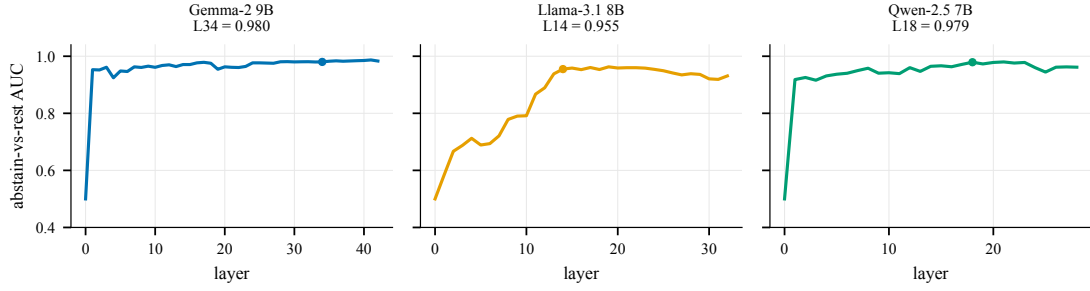


Figure 1: Cross-family replication of the cautious-framing abstention probe. Per-layer abstain-versus-rest AUC for Gemma-2-9B-IT, Llama-3.1-8B-Instruct, and Qwen-2.5-7B-Instruct, with the best-layer marker. Same-layer cosines to the refusal direction are reported in Table 1.

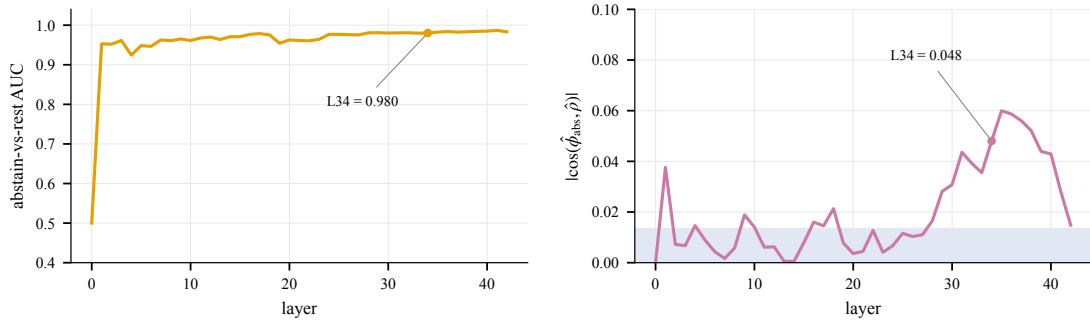


Figure 2: Gemma abstention is decodable and geometrically dissociated from refusal. Left, cautious-only abstain-versus-rest probe AUC by layer, peaking at 0.980 at layer 34. Right, $|\cos|$ between the abstain probe direction and the refusal direction by layer, with the isotropic-null band at $\mathbb{E}[|\cos|] = \sqrt{2/(\pi d)} \approx 0.013$. The same-layer cosine at the abstain peak is 0.048, within a factor of four of the null and far below any value implying a shared direction.

knows probe (Kadavath et al., 2022; Yin et al., 2023; Slobodkin et al., 2023), since the question-only correctness axis shows low cosine alignment with the abstain direction across all three families. It is also not the conflict-detection direction of Zhao et al. (2024), whose probe separates conflict-present from conflict-absent prompts whereas ours is trained inside conflict-present prompts and separates abstention from PK-win and CK-win.

The recipe behind the dissociation is not a guardrail. A single mean-difference vector at one layer is a blunt instrument for a feature that probably emerges across many layers, and PopQA’s templated conflicts can yield a direction that is template-specific even when the underlying feature is general. The dissociation does, however, point to two follow-ups. Abstention can be extracted jointly across layers rather than at a single layer, and template counterfactuals can be replaced with naturalistic conflicts to test whether the direction is template-specific.

Limitations

The intervention fails the factual-loss gate and does not transfer to SciQ or NQ-open, so the abstain-direction recipe is PopQA-specific even when the underlying decodable feature is not. The regex-and-alias label under-recovers true abstentions against model-judge audits, so absolute rates are conservative while paired causal deltas are unaffected, and the probe is pooled across all four context cells rather than the counterfactual cell alone (Appendix A). The cross-intervention rests on one refusal extraction (Arditi et al., 2024) and one abstain extraction at a single layer and scale, so the dissociation is established against the canonical refusal extractor rather than against the class of refusal-shaped directions.

The result characterises abstention under a known fact with conflicting context rather than under genuine ignorance, template counterfactuals can overstate context-utilisation effects (Hagström et al., 2025), and over-abstention itself remains a risk in medical, legal, and policy settings (Perez et al., 2023; Sharma et al., 2024).

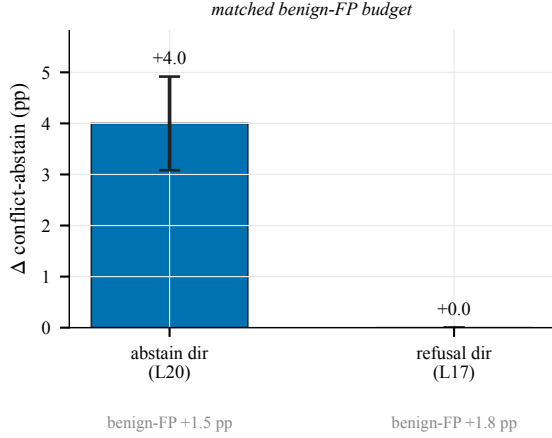


Figure 3: Matched-budget steering comparison on Gemma at scale $s = +2$. Abstain-direction steering raises conflict abstention by +4.0 pp at +1.5 pp benign-FP cost, while refusal-direction steering produces zero conflict-abstain shift at a comparable +1.8 pp benign-FP cost. Error bars are normal-approximation 95% CIs over five seeds.

Ethical considerations

JailbreakBench prompts are used only to compute a refusal direction. No harmful generations are released. The reported abstain-steering recipe fails its factual-loss gate, so we frame it as a mechanistic characterization rather than a deployable filter.

AI Usage Disclosure

Large language models assisted with code changes, experiment orchestration, artifact checking, figure regeneration, copy-editing, reviewer simulation, manuscript restructuring, and transcription of script-computed numerical results into the manuscript. All reported experimental numbers were produced by the analysis scripts from saved run artifacts and checked against the corresponding `metrics.json` files; no numerical value is reported from an unaudited model assertion.

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A Cell Composition of the Cautious-Framing Probe

Table 4 reports per-cell abstain counts in the cautious-framing subset (946 rows). Table 5 reports three probe variants, namely the pooled probe trained on all four cells, the same probe trained without the irrelevant-context cell, and a counterfactual-only probe.

Cell	Abstain count
pk_known $\times$ ck_irrelevant	215
pk_known $\times$ no_ck	26
pk_known $\times$ ck_false	0
pk_known $\times$ ck_true	0

Table 4: Per-cell abstain counts in the cautious-framing subset. Most cautious abstentions arise under irrelevant context. The counterfactual cell has zero abstain labels, so a counterfactual-only probe is undefined.

Variant	Filter	Test AUC
(A) pooled cautious	all four cells	0.980 (L34)
(B) no irrelevant	exclude ck_irrelevant	0.973 (L22)
(C) ck_false only	ck_false only	N/A

Table 5: Probe variants on the cautious-framing subset. Variant (A) replicates the headline result. Variant (B) shows the signal survives when irrelevant-context rows are dropped. Variant (C) is undefined because the abstain class is empty in the counterfactual cell.

B Reproducibility and Infrastructure

Hardware. The rerun uses three NVIDIA B200 GPUs (180 GB HBM3e each) on a Linux host with CUDA 13.0 and driver 595.71.05. Model inference uses bfloat16 (bf16) weights and activations. Probe training uses float32 (fp32). We enable expandable PyTorch CUDA segments through the

standard `PYTORCH_CUDA_ALLOC_CONF` environment variable for long sweeps. Flash-attention is unavailable on the host, so attention falls back to PyTorch scaled dot-product attention (SDPA).

Software. Python 3.12.3 managed by `uv`, PyTorch (Paszke et al., 2019) 2.11.0+cu130, transformers 5.8.1, accelerate 1.13.0, scikit-learn 1.8.0, sae_lens 6.43.0. The full environment is locked in `uv.lock`.

Code and artifacts. Code, configuration files, per-run `metrics.json` files, the two-judge audit CSV, and the small direction vectors used in the steering experiments will be released upon acceptance.

Models and weights. We use three HuggingFace checkpoints at the following pinned revisions.

```
google/gemma-2-9b-it at 11c9b30
meta-llama/Llama-3.1-8B-Instruct
at 0e9e39f
Qwen/Qwen2.5-7B-Instruct
at a09a354
```

Seeds. Stochastic generation and steering entry points accept `-seed` and use the project seeding utility to fix `random`, `numpy`, and `torch` seeds on CPU and CUDA. Five-seed steering uses seeds 0–4. Three-seed scope checks use seeds 0–2. Aggregation scripts are deterministic reductions of those saved runs.

Run-directory contract. Core config-driven stages copy their YAML configuration into the run directory; diagnostic and aggregation stages record command-specific metadata instead. Reported stages write `meta.json` (including git SHA, hostname, GPU, and package versions) and `metrics.json`, with larger tensor or row-level artifacts stored separately as `.npz`, `.parquet`, or `.csv`. Probe AUCs, cosines, and steering deltas reported in the paper are read from these `metrics.json` files.

Compute budget. The full rerun (data build, residual extraction across three models, per-layer probe training, refusal-direction extraction, five-seed steering on PopQA, three-seed scope checks on SciQ and NQ-open, and two-judge label audit on 800 rows) ran on three B200s in parallel in approximately 25 minutes wall time, or 1.2 allocated GPU-hours.